

CetaSignal

A Constraint-Based Acoustic Classifier for Cetacean Coordination Signals

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CetaSignal is a rule-based acoustic classification framework designed to infer whale behavioral coordination states from vocalization cadence features. Building on foundational work in marine bioacoustics by Payne & McVay (1971), Tyack & Clark (2000), Clark (2005), Whitehead (2003), and Oestreich et al. (2020, 2022, 2024), the system interprets acoustic signals as coordination protocols shaped by the physical constraints of the ocean environment. The five-class behavioral coordination taxonomy — Contact, Dive, Forage, Navigate, and Broadcast — is an original unified framework proposed here, derived from ocean physics and cetacean ecological constraints. Individual behavioral-acoustic relationships documented by Oestreich and colleagues provide direct empirical validation for the Forage and Navigate classes. The classifier uses transparent, traceable decision rules rather than opaque machine learning. The current prototype evaluates ~5,000 specimens across 19 cetacean species.

1. THE CONSTRAINT FRAMEWORK

Cetacean acoustic communication evolved under the unique physical constraints of the ocean environment. Tyack & Clark (2000) established that the ocean presents a fundamentally different sensory landscape than terrestrial environments: visual range is severely limited (rarely exceeding tens of meters even in clear water), while sound propagates with exceptional efficiency [1]. These two properties together create the primary selective pressure for acoustic signaling in cetaceans.

Low Visibility

Coastal waters often provide visibility of only a few meters. Because cetacean pods may operate at spatial separations far exceeding these distances, visual communication is insufficient for maintaining group cohesion — driving the evolution of acoustic coordination mechanisms [1].

Three-Dimensional Spatial Coordination

Unlike terrestrial animals operating primarily in two dimensions, cetaceans must coordinate movement through a fully volumetric environment. Depth changes, heading adjustments, and lateral spacing occur simultaneously. Effective coordination signals must therefore encode directional and temporal information relevant to this three-dimensional context [1].

The SOFAR Channel and Long-Range Propagation

The SOFAR (Sound Fixing and Ranging) channel acts as a natural acoustic waveguide in the deep ocean, allowing low-frequency vocalizations to travel thousands of kilometers with relatively low attenuation. Christopher Clark's foundational work at Cornell's Bioacoustics Research Program — using the U.S. Navy SOSUS hydrophone network — demonstrated that blue, fin, and humpback whale calls can be detected across ocean-basin scales, and that coherent migration of multiple individuals can be tracked acoustically in near real-time [2]. This physical property creates a strong evolutionary incentive for acoustic signaling systems optimized for:

- Long-range propagation with minimal signal degradation
- Reliable detection against ambient ocean noise
- Rapid coordination across spatially distributed pod members

Frequency-Dependent Attenuation

Higher frequency sounds attenuate far more rapidly in seawater than low-frequency signals. Different frequency ranges therefore correspond to different effective communication radii. Low-frequency calls (below ~200 Hz) propagate across basin scales and are associated with navigation and broadcast functions; higher frequencies provide spatial resolution for short-range coordination or echolocation [1, 2].

Core Hypothesis

These physical constraints lead to the central hypothesis of CetaSignal: cetacean vocalizations evolved primarily not as symbolic language units, but as behavioral coordination protocols shaped by the acoustic ecology of the marine environment. Vocal signals serve to reduce uncertainty among distributed individuals by communicating behavioral state transitions. This framing aligns with Tyack & Clark's (2000) functional interpretation of cetacean acoustic behavior [1], and with Whitehead's (2003) model of how ocean environment shapes social behavior in deep-diving odontocetes [3]. Coordination problems solved by such signals include:

- Maintaining pod cohesion across visual range
- Initiating and synchronizing dives
- Coordinating foraging maneuvers
- Synchronizing migration heading
- Advertising presence across large distances

2. ACOUSTIC FEATURE EXTRACTION

The classifier relies on six acoustic features measurable directly from recorded signals. These features were selected because they are physically interpretable, extractable from standard spectrographic analysis, and have documented relationships to behavioral contexts in the marine bioacoustics literature.

Peak Frequency

The dominant frequency of the signal within a given time window, measured using Fast Fourier Transform (FFT) spectral analysis. Corresponds to the frequency bin with the highest power. Peak frequency is a primary indicator of effective communication range, given frequency-dependent attenuation in seawater [1].

Frequency Contour

How the dominant frequency changes over time, tracked across successive spectrogram frames. Categorized as rising, descending, flat, or complex. North Atlantic right whale upcalls, for example, exhibit rising contours spanning approximately 50–400 Hz and lasting 1–2 seconds, consistent with contact and cohesion functions [4, 5].

Call Duration

The time between signal onset and offset, determined by amplitude thresholds or energy-based detection. Duration differentiates signal classes: short bursts are associated with dive initiation, while extended vocalizations characterize navigation and broadcast signals.

Inter-Pulse Interval (IPI)

For pulsed signals, the time spacing between successive pulses. Particularly relevant for odontocete click trains and sperm whale codas, which Whitehead, Rendell, and colleagues have shown carry individual and clan-level identity information [3, 6].

Repetition Rate

The frequency of call occurrence within a given time window, expressed in calls per minute. High repetition rates are associated with active social coordination or prey pursuit; slow repetition characterizes navigation signals and broadcast songs [1].

Urgency Index

A derived feature approximating information density within a signal: *Urgency Index* = $(\text{Pulse Density} \times \text{Repetition Rate}) / \text{Signal Duration}$. Elevated urgency index values correspond to rapid behavioral transitions such as prey pursuit or predator avoidance.

3. BEHAVIORAL COORDINATION CLASSES

The five behavioral coordination classes below constitute an original unified taxonomy proposed by this framework. While individual behavioral-acoustic relationships have been documented across the literature, no prior work has organized cetacean coordination signals into a single constraint-derived classification scheme spanning Contact, Dive, Forage, Navigate, and Broadcast functions. The taxonomy is grounded in the physical constraints of Section 1 and in the functional communication model of Tyack & Clark (2000) [1].

Critically, Oestreich et al. (2020) provided the most direct empirical support for acoustic behavioral classification to date: combining MBARI hydrophone recordings with biologging tags on individual blue whales, they demonstrated that diel patterning of blue whale songs encodes foraging vs. southward

breeding migration behavioral states — validating the core premise underlying the Forage and Navigate classes [9]. Oestreich et al. (2024) further argued that behavioral bioacoustics — decoding acoustic signals to infer behavioral state — is broadly underused and requires exactly the kind of structured classification framework proposed here [11].

Class	Primary Function	Typical Acoustic Features
Contact	Pod cohesion; convergence between separated individuals	Rising contour, 100–400 Hz, moderate duration (0.5–2 s), low urgency
Dive	Initiation and synchronization of descent	Descending contour, short duration (<1 s), rapid onset
Forage	Prey detection and hunting coordination	High-frequency pulses or click trains, elevated urgency index [9]
Navigate	Sustained migration heading alignment	Low frequency (<200 Hz), flat contour, slow repetition, long duration [9, 10]
Broadcast	Long-range advertisement or reproductive display	Complex/hierarchical structure, very low frequency, extended sequences [7]

4. EXAMPLE CASE STUDY

The following case illustrates the classifier's decision logic using North Atlantic right whale upcall recordings. The upcall is among the most extensively documented cetacean contact signals in the literature, first characterized systematically by Clark (1982) and subsequently studied in detail by Parks & Tyack (2005) and Parks, Urazghildiiev & Clark (2009) [4, 5].

Parameter	Measurement
Species	North Atlantic Right Whale (<i>Eubalaena glacialis</i>)
Signal Type	Upcall (stereotyped tonal upsweep)
Peak Frequency	~170 Hz (fundamental)
Frequency Contour	Rising (~100 Hz to ~200 Hz)
Call Duration	~1.2 seconds
Inter-Pulse Interval	None detected (tonal, not pulsed)
Urgency Index	Low (isolated call, non-burst)
CetaSignal Classification	CONTACT — pod cohesion / convergence signal

Independent behavioral observations associated with the recording indicate that two whales initially separated by approximately 200 meters converged to within roughly 50 meters following the exchange of upcalls. This convergence behavior is consistent with a contact or cohesion function. The classification aligns with the behavioral interpretation established by Parks & Tyack (2005), who documented that upcalls are produced by all age classes and sexes throughout the species' range and function as the primary contact signal [4].

5. DATASET AND EVALUATION

The current prototype evaluates approximately 5,000 acoustic specimens drawn from publicly available marine bioacoustic datasets and curated hydrophone archives, spanning 19 cetacean species across multiple ocean basins. Datasets include publicly archived recordings from institutions including the Cornell Lab of Ornithology Macaulay Library, NOAA passive acoustic monitoring archives, and the Watkins Marine Mammal Sound Database.

Each specimen in the evaluation dataset includes:

- A recorded acoustic signal with associated metadata
- Extracted acoustic features (the only inputs to the classifier)
- An independently documented behavioral annotation drawn from concurrent observational records

During evaluation, the classifier receives only acoustic features and produces a behavioral class prediction. This prediction is then compared against the independently established behavioral record associated with the same recording. Because the classifier is rule-based, each classification decision is fully traceable to specific acoustic features and their mapped decision rules — a transparency property not available in learned neural classifiers.

Performance metrics include overall classification accuracy, class-specific recall and precision, and confusion matrices across all five behavioral classes.

6. FUTURE RESEARCH DIRECTIONS

Oestreich et al. (2024) identified behavioral bioacoustics — using acoustic signals to infer behavioral state — as an emerging field that is critically underused relative to its potential [11]. CetaSignal is positioned directly within that gap. The following directions would advance both the classifier's empirical grounding and its contribution to the broader field.

Integration with Tag Data

Combining acoustic recordings with high-resolution biologging data from suction-cup-attached DTAGs (Digital Acoustic Tags, as used by Tyack and colleagues) would provide stronger behavioral validation by directly correlating vocal signals with simultaneous depth, heading, and movement profiles [1].

Controlled Playback Experiments

Playback experiments could test whether specific acoustic signal structures reliably trigger predicted behavioral responses in target individuals — a methodological approach with precedent in the cetacean communication literature.

Cross-Species Analysis

Comparative analysis across the 19 species in the current dataset could determine whether coordination signal acoustic structures are conserved across cetacean taxa or highly species-specific. Whitehead & Rendell's work on sperm whale vocal clan dialects suggests that even within-species variation encodes social structure [3, 6].

Juvenile Signal Acquisition

Understanding how juvenile whales acquire and refine coordination signals during development may provide insight into the learning mechanisms and cultural transmission pathways of these protocols. Root-Gutteridge et al. (2018) documented lifelong acoustic maturation in North Atlantic right whales, suggesting that signal refinement is an extended developmental process [8].

Anthropogenic Noise Impact

Clark and colleagues have documented that shipping noise has reduced the effective communication space of North Atlantic right whales by up to 97% in some habitats [2, 5]. Future work could model how degraded acoustic environments affect classifier performance, and by extension, the effectiveness of natural coordination signaling.

7. CONCLUSION

CetaSignal represents an exploratory research prototype investigating how acoustic signal structure relates to behavioral coordination in cetaceans. The five-class behavioral coordination taxonomy — Contact, Dive, Forage, Navigate, Broadcast — is an original unified framework proposed here, derived from the physical and ecological constraints governing cetacean communication. The framework is grounded in established marine bioacoustics literature — particularly Tyack & Clark (2000) on communication function [1], Clark (2005) on basin-scale monitoring [2], Payne & McVay (1971) on broadcast signal structure [7], and Whitehead (2003) on social coordination [3]. The Forage and Navigate classes draw direct empirical validation from Oestreich et al. (2020), which demonstrated that blue whale acoustic signatures encode foraging vs. migration behavioral states at both individual and population scale [9].

Oestreich et al. (2024) identified behavioral bioacoustics as critically underused relative to its potential for understanding animal responses to ecosystem change [11]. CetaSignal is a direct response to that gap: a constraint-based, interpretable classification framework designed to decode coordination function from acoustic structure across species and ocean basins.

Further empirical validation and collaboration with marine bioacoustics researchers — particularly those with access to concurrent behavioral and acoustic data streams — would meaningfully advance the system's predictive accuracy and theoretical grounding. We welcome engagement on shared datasets,

playback methodology, and cross-species analysis.

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